Exercise: block-coordinate descent for a least-squares problem

Let $y \in \mathbb{R}^3$ and

$$f(x) = \frac{1}{2}\|Ax - y\|^2, \qquad A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a_1 & a_2 \end{pmatrix}, \qquad x = (x_1, x_2) \in \mathbb{R}^2.$$

1. Compute $\nabla f(x)$, $\nabla_1 f(x)$ and $\nabla_2 f(x)$, then the Lipschitz constant β of ∇f and the coordinate-wise constants L_1 and L_2.
2. Write the coordinate descent update of x_1 with step size $1/L_1$. Show that it exactly minimizes f with respect to x_1, for fixed x_2.